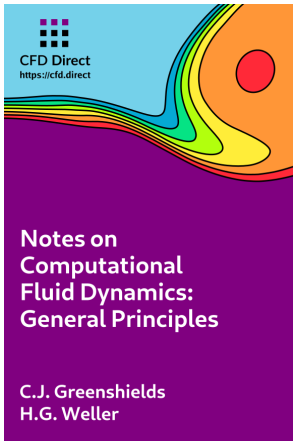
Notes on Computational Fluid Dynamics: General Principles



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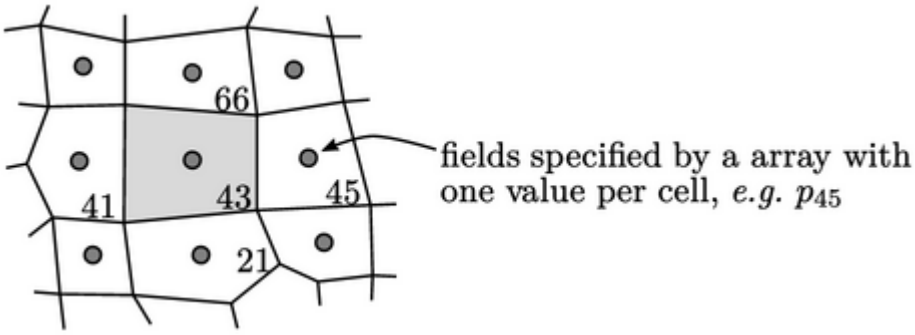
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3.4 Equation discretisation

Equation discretisation converts partial differential equations for *continuous* fields, *e.g.* pressure p, into sets of linear equations for *discrete* fields.

The values of the principal fields, *e.g.* p, are associated with cells. A field is then represented by an array of values, p_i, for cell indices $i = 1, 2, 3, \ldots, N$. N is the total number of cells.



Equation discretisation creates a linear equation for each cell. For cell 43 above, the equation might have the following form:

$$a_{43,21}\,p_{21} + a_{43,41}\,p_{41} + \mathbf{a_{43,43}\,p_{43}} + a_{43,45}\,p_{45} + a_{43,66}\,p_{66} = b_{43}.$$

where $a_{i,j}$ and b_i are coefficients corresponding to cell indices i,j (diagonal coefficient in **bold**). The set of linear equations for all cells can be written as a matrix equation of the form:

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,N} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,N} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{N,1} & a_{N,2} & a_{N,3} & \cdots & a_{N,N} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ \vdots \\ p_N \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_N \end{bmatrix} \tag{3.1}$$

The matrix contains of a set of coefficients $a_{i,j}$ where each row i corresponds to the linear equation for the cell with index i. Each row of coefficients are non-zero only for the respective cell (diagonal $a_{i,i}$) and near-neighbours. All other coefficients are zero, making the matrix extremely *sparse*.

The matrix equation can be represented as $[\mathbf{A}][p] = [\mathbf{b}]$, where: $[\mathbf{A}]$ are the matrix coefficients; $[\mathbf{b}]$ are the source coefficients b_i; and, $[p]$ is the discretised pressure field. It may also be illustrated as follows, where '$\star$' indicates non-zero coefficients.

$$\begin{bmatrix} \star & \star & & \star \\ \star & \star & \star & \star \\ & \star & \star & \star \\ \star & & \star & \star \end{bmatrix} \begin{bmatrix} p \end{bmatrix} = \begin{bmatrix} \star \\ \star \\ \star \\ \star \end{bmatrix}$$

Segregated solution

A CFD simulation generally solves a set of physical equations, *e.g.* for mass, momentum and energy conservation. The finite volume method traditionally discretises each physical equation separately to form individual matrix equations for single solution variables, *e.g.* $[p]$, $[\mathbf{u}]$, $[T]$, rather than creating a single matrix equation that represents all the physical equations.

The *segregated* matrix equations are solved one variable at a time, *e.g.* solving for $[p]$, $[\mathbf{u}]$ and $[T]$ in separate steps. Where the solution variable is a vector or tensor, *e.g.* $\mathbf{u}$, it is *decoupled* into individual matrix equations for each component, *e.g.* $[u_x]$, $[u_y]$, $[u_z]$.

The matrix equations are solved in an *iterative* sequence, in which the equation for one variable, *e.g.* $[u_x]$, incorporates current values of other variables, *e.g.* u_y, u_z and p, into the source vector, as shown below.

$$\begin{bmatrix} \star & \star & & \star \\ \star & \star & \star & \star \\ & \star & \star & \star \\ \star & & \star & \star \end{bmatrix} \begin{bmatrix} u_x \end{bmatrix} = \begin{bmatrix} \star \\ \star \\ \star \\ \star \end{bmatrix} \leftarrow \text{includes known } u_y,\ u_z,\ p$$

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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